

Design Document:

Project Title: Bike Crash Detection System Using Machine Learning

1. System Overview

The proposed system is an intelligent bike crash detection system that uses sensor data and a hybrid CNN-LSTM deep learning model to detect accidents in real time. The system collects motion data from accelerometers and gyroscopes, processes the data, and identifies crash events.

2. System Architecture

The system consists of the following modules:

Sensor Module

- Collects acceleration and gyroscope data.
- Monitors bike movement continuously.

Data Processing Module

- Removes noise from sensor data.
- Performs normalization and preprocessing.

Machine Learning Module

- Uses CNN for feature extraction.
- Uses LSTM for analyzing time-based motion patterns.
- Classifies events as crash or normal riding.

Alert Module

- Generates emergency alerts.
- Sends crash location and time information.

3. Data Flow

Input:

Sensor Data → Preprocessing → Feature Extraction → CNN-LSTM Model → Crash Prediction → Alert Generation

4. Modules Description

Data Acquisition Module

Collects real-time motion data using sensors.

Preprocessing Module

Improves data quality using filtering and normalization techniques.

Crash Detection Module

Analyzes sensor patterns and detects abnormal events.

Emergency Alert Module

Sends notifications when a crash is detected.

5. Technologies Used

Software:

- Python
- TensorFlow
- Keras
- NumPy
- Pandas

Hardware:

- Accelerometer
- Gyroscope
- GPS Module

6. Algorithm Used

Hybrid CNN-LSTM Model:

CNN:

- Extracts important spatial patterns from sensor data.

LSTM:

- Captures sequential changes in motion data.

The combined model improves accuracy and reduces false detection.

7. System Requirements

Hardware:

- Smartphone/Embedded Device
- Sensors
- GPS

Software:

- Python Environment
- Machine Learning Libraries

8. Conclusion

The design provides an efficient crash detection solution with real-time monitoring and emergency alert generation.